Plastic vs. Planet

A 30-second AI-generated short film comparing plastic packaging with banana-leaf packaging

Hackathon 4: Picture the Planet · SDG 13 Climate Action · Made with ComfyUI

Milou van Leuven & Kadir Guclu · AI for Good, De Haagse Hogeschool



Watch the film first

30 seconds · 8 shots · every frame generated in ComfyUI

https://youtu.be/Y_W9cMousSE?is=wEJwg0T4H8UXx_pR

The problem: plastic packaging creates a huge amount of waste

35.3 kg

plastic packaging waste per person in the EU (2023)

14.8 kg

of that was recycled: only about 42%

430 Mt

of plastic produced worldwide every year

36%

of all plastic production is used for packaging



Why this is a climate problem (SDG 13): plastic is made from fossil fuels and is often burned after use. Over its life cycle, plastic caused 3.4% of global greenhouse gas emissions in 2019.

For whom: everyday consumers who buy plastic-packaged products and rarely see what happens after they throw it away.

Sources: Eurostat, plastic packaging waste statistics (2023 data); UNEP, global plastic production; OECD (2022), Global Plastics Outlook.

Our audience

Everyday consumers who buy plastic-packaged products

What they already believe

- Plastic is bad, mainly for sea animals
- Most of what they throw away gets recycled
- Packaging is a small detail next to flying or meat

Who the film is not for

- Policy makers and packaging experts: they need data, not a 30-second film
- Young children: the waste and pollution scenes are too heavy
- People who already shop zero-waste

The concept: two lives of one package

	Production	Supermarket	After use	Nature
Plastic	Dark factory, smoke, mountains of plastic	Every item wrapped in its own plastic	Thrown away: landfill and incinerator smoke	Drifts in the sea, entangles and is eaten by animals
Banana leaf	Sunny banana plantation, leaves harvested and folded into packaging	The same shelf, products wrapped in banana leaf	Leaf tossed onto a garden compost heap	Breaks down, eaten by animals, returns to the soil

Shot list: 8 shots, about 3.5 seconds each

1

Production · plastic

Dark plastic factory, machines, smoke, heaps of plastic

2

Production · leaf

Green banana plantation in sunlight, leaves harvested and made into packaging

3

Supermarket · plastic

Shelf where fruit, vegetables and food are each wrapped in plastic

4

Supermarket · leaf

Same shelf, products wrapped in banana leaf, a much greener image

5

After use · plastic

Plastic thrown away: huge waste pile, incinerator with smoke

6

After use · leaf

Banana leaf thrown onto a compost heap among plants and soil

7

Nature · plastic

Plastic floats in the sea, animals get tangled or eat it

8

Nature · leaf

Banana leaf in nature breaks down, is eaten, disappears into the soil

The ComfyUI workflow: shot list in, film out

1

Prompt

Positive + negative prompt per shot

›

2

Checkpoint

Stable Diffusion model, [model name]

›

3

KSampler

Fixed seed and settings per shot

›

4

Keyframe

One still image per shot

›

5

Video node

[AnimateDiff / SVD / Wan / LTX] adds motion

›

6

Edit

Cut the 8 clips, add sound and end card



Reproducible: the workflow JSON, the seed and the settings for every shot are in our GitHub repo, so anyone can regenerate the exact film.

Why this film reaches our audience



It makes the invisible visible

Our audience thinks of plastic as an ocean problem. Showing the factory and the incinerator puts the climate cost on screen.



It ends with an option, not guilt

Every plastic shot is answered by a leaf shot. The viewer leaves with a picture of an alternative, which motivates more than fear alone.



It speaks at the moment of choice

Consumers choose packaging every week in the supermarket. A 30-second film fits how people watch content on social media.

Ethical reflection

Oversimplifying

Banana leaf is not a solution everywhere: it has to be grown and shipped too. The end card frames it as one example of natural packaging.

Faces and places

No real people, brands or recognisable locations appear. Factory, shop and beach are fictional.

Is it real?

An "AI-generated" label is visible during the film, so nobody mistakes the scenes for documentary footage.

What we learned

- Fixing the seed per shot is what makes a generated film reproducible.
- Contrast pairs tell the story faster than text on screen.
- Generating motion is slow and costly, so planning the shot list first saved many generations.

Thank you. Questions?